

LM in Artificial Intelligence
Image Processing and Computer Vision
17/07/2025

- 1) (A) Define the concept of scale invariance for local features, then describe the scale-space, why it has been introduced and what tool can be used to compute it; (B) Describe the Lindeberg salient point features detector, its mathematical formulation, what type of features it detects and how a salient point is associated with a specific scale; (C) Finally, briefly describe the relation between the Lindeberg detector and the Difference of Gaussian detector proposed by Lowe.
- 2) (A) Describe in detail the motivating experiment for the introduction of Residual Networks (ResNets) and discuss if it shows underfitting or overfitting, justifying your answer and explaining why the architecture ResNets should reduce it. (B) Draw the standard and bottleneck residual blocks used in ResNets and compute their numbers of parameters and FLOPs. Compare them to justify the introduction of the bottleneck variant. (C) Finally, discuss why and how the standard residual block is modified at the beginning of a stage.